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Rana Abubakar Khan

If you want to learn computer programming then contact with me

truefriendlion@gmail.com

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www.facebook.com/groups/vu1234/

PHY301 MID TERM PAPER SHARED BY STUDENT

ON JANUARY 14, 2015 AT 8:33PM

PHY301 Current Mid Term Papers PHY301 Sample Midterm Papers Pattern:

Paper Pattern

Total Questions: 26 Total Marks: 40

Total MCQs of 1 Mark: 20

Short Subjective Question of 2 Marks: 2

Subjective Question of 3 Marks: 2

Subjective Questions of 5 Marks: 2

Syllabus to be covered: Lecture 1-22.

Shared By VU Students:

Total Marks: 40



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Total Questions: 26

Total MCQs of 1 Mark: 20

Short Subjective Question of 2 Marks: 2

Subjective Question of 3 Marks: 2

Subjective Questions of 5 Marks: 2

1: current divider rule se circuit solve krna tha

2: aik question main circuit given ha uss ka mesh 1 and 2 ki kvl equation btani thi

3: circuit given tha KVL equation write krni meah 1 and 2 ki thi

4: matrix equation se solve krna tha aik question 5 marks ka

What is the resistance level of photodiode in total darkness? (2 Marks)

Write constraint equation for given circuit? (2 Marks)

What is band gap energy? (2 Marks)

Determine the current in secondary (about transformer)? (3 Marks)

Write KVL equation for outer loop? (3 Marks)

For the circuit shown in the figure using ideal diode, find the value of the voltage and current? (3 Marks)

Determine the DC load current for the rectifier shown in fig.? (5 Marks)

Find V_{10} by division Rule? (5 Marks)

How we will convert 10V voltage source into current in flowing circuit, what will be the value of current source? (5 Marks)

PHY301 MID TERM PAPER SHARED BY STUDENT

ON JANUARY 14, 2015 AT 8:33PM

My Today's Paper

Shared By Fariha Sister (VU STUDENT):

1. MCQs almost sab past paper wali file se aaye thay.
2. Subjective mein pooray 3 question kvl equation k aaye.
3. Kirshoff' Voltage Law equation walay k write kvl equation for the given circuit.
4. Aik question easy sa tha k given circuit k kitnay loops ban saktay hein.
5. Aur remaining 2 thay jin mein circuits diye huay thay aur I_3 ya I_4 find karna tha



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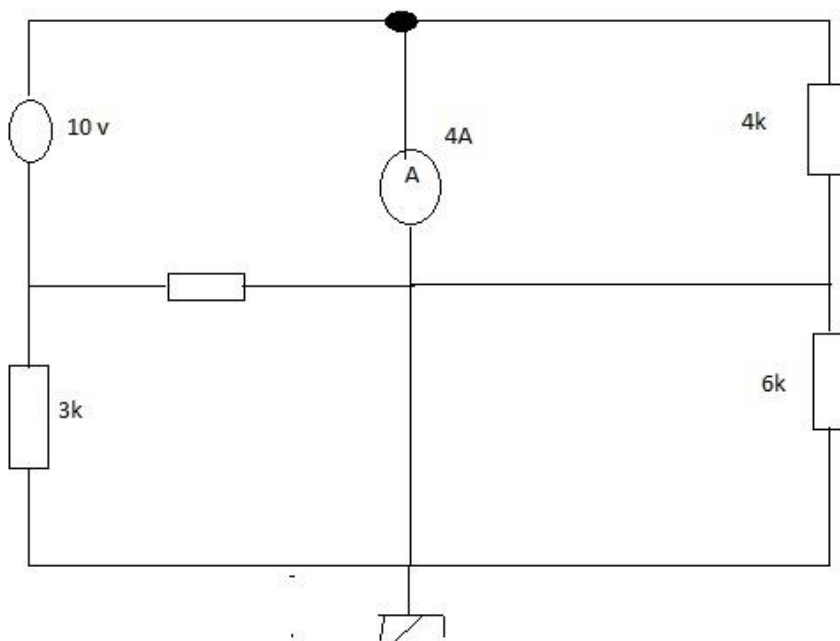
PHY301 MID TERM PAPER SHARED BY STUDENT

ON JANUARY 16, 2015 AT 7:00PM

My Today's PHY301 Paper

Label the Node and write super node equation

5Marks



2- 1 aur given circuit tha us ma path adcba find karna tha

3- find I_o kvl identify and label each mesh (is ka b circuit dia huwa tha)

4- Calculate V_{ac} using KVL (is ka b circuit dia huwa tha)

5- 1 question ma Kirchhoff 's law ki madad sa equation banani thi.



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PHY301 MID TERM PAPER SHARED BY STUDENT

ON DECEMBER 21, 2013 AT 7:57AM

Total Questions: 26 Total Marks: 40

Total MCQs of 1 Mark: 20

Short Subjective Question of 2 Marks: 2

Subjective Question of 3 Marks: 2 Subjective Questions of 5 Marks: 2

1:current divider rule se circuit solve krna tha

2:aik question main circuit given ha uss ka mesh 1 and 2 ki kvl equation bnti thi

3:circuit given tha KVL equation write krni mesh 1 and 2 ki thi

4:matrix equation se solve krna tha aik question 5 marks ka

PHY301 MID TERM PAPER SHARED BY STUDENT

Sunday DATE 16:12:2012

PHY 301

SOLVE MID TERM PAPER

MY TODAY PAPER

Paper time 4:00 pm



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Question #1

If we connect inductance in series total inductance will be

1: sum of individual inductance

2: reciprocal of combined effect of all these inductance

3: product of all these

4: sum of 1st and last inductance

Question #2

Distance or link b/w two nodes is called

1: branch

2: loop

3: node

4: super mesh

Question #3



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Constraint or coupling equation describes

- 1: simple loop
- 2: super node mathematically
- 3: ground
- 4: simple node

Question #4

Which of the following statement is not true?

- 1: An open circuit has zero current
- 2: A short circuit has excessive current
- 3: An open circuit and a short circuit have opposite effect on resistance and current
- 4: Close circuits don't allow flow of current

Question #5

When two resistances are connected in series

- 1: they must both have same resistance value
- 2: the voltage across each must be the same
- 3: They must have different resistance value
- 4: There is only one path for current for both resistances

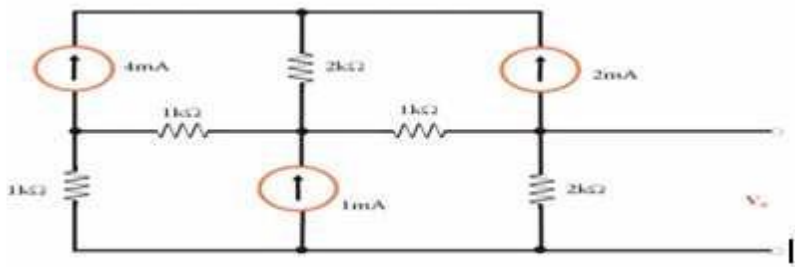


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Question #6

Which current source will be used for super mesh?



1: 4mA

2: 2mA

3: 1mA when two meshes share current source then it became super

Mesh

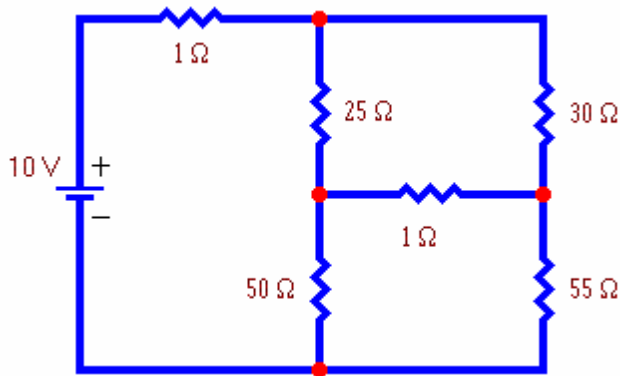
4: 2mA and 4mA

Question #7

How many KVL equations can be written for the given circuit?



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1: 2

2: 4

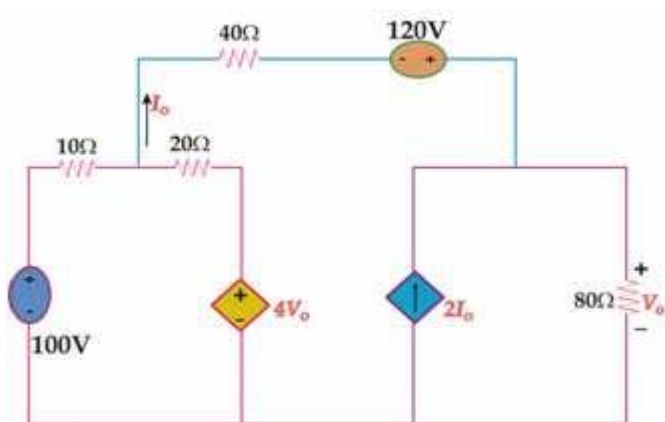
3: 3 because 3 meshes or loops are formed and number of loops is

Equal to number of equations

4: 1

Question #8

How many sources are dependent sources in given circuit?



1: 3



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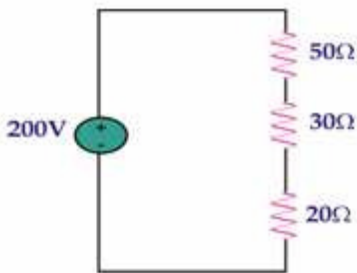
2: 2

3: 4

4: 1

Question #9

Through which resistance most current will flow



1: 50 ohm

2: 30 ohm

3: 20 ohm

4: same through all resistance because same current flowing through

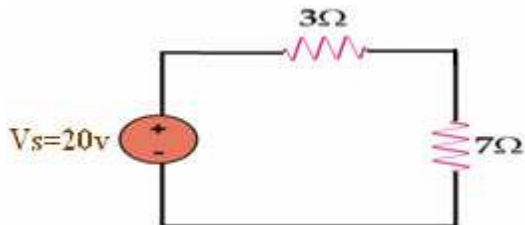
Series combination Page 32

Question #10

In the figure below Voltage drop across 7ohm will be



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1: 20v

2: 14v

$$V = R/RT * VS$$

So,

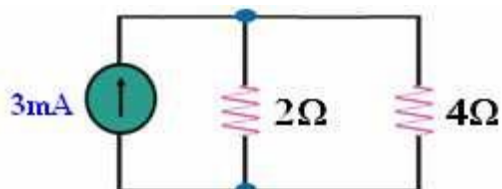
$$V = 7/10 * 20 = 14V$$

3: 6v

4: 10v

Question #11

If current flowing through 4 ohm is 1Ma, voltage drop across it will be



1: 2V

2: 6V



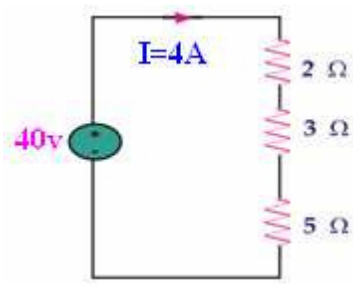
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3: 4V

4: 12V

Question #12

For the given figure current flowing through 5ohm is



1: 5A

2: 4A

3: 10A

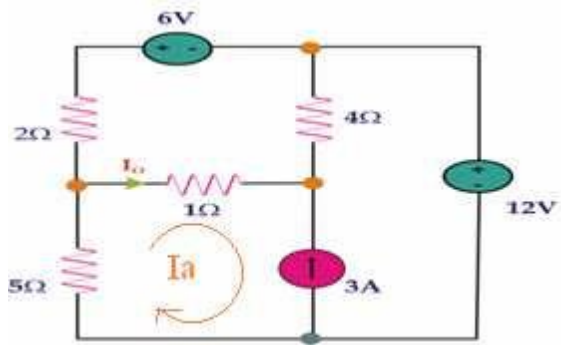
4: 2A

Question # 13

The value of I_a for given circuit is



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1: $I_a = I_o$

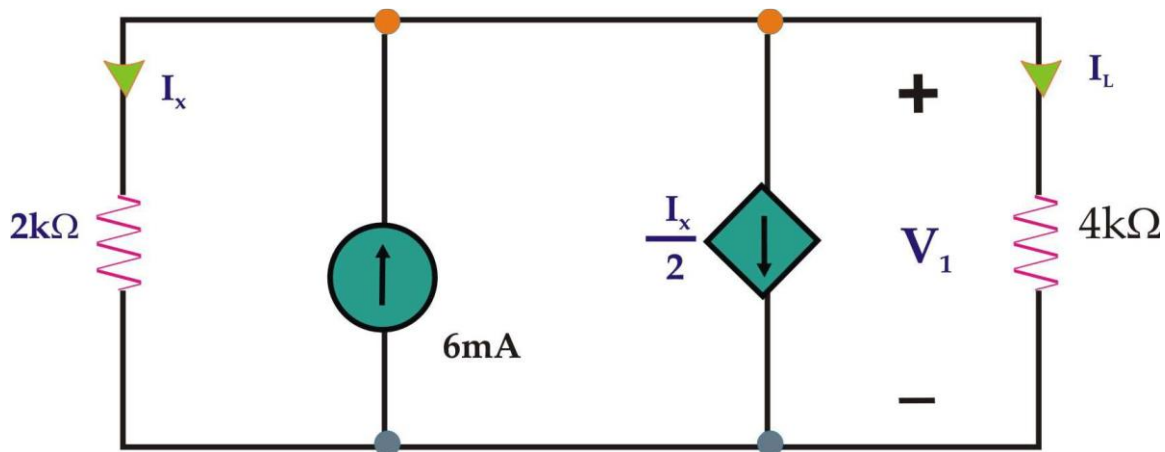
2: $I_a = 3A$

3: $I_a = -3A$ Current is not in same direction with 3A.

4: $I_a = 6V$

Question # 14

In the given circuit the value of independent current source is





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1: I_x

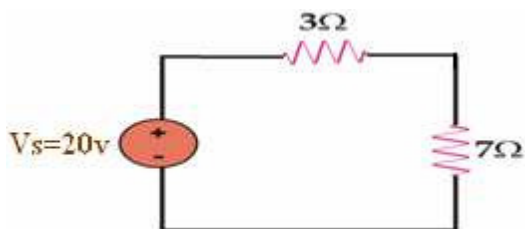
2: $I_x/2$

3: 6Ma

4: I_L

Question # 15

Using voltage divider in the figure below Voltage drop across 3ohm will be



1: 20v

2: 14v

3: 6v

$$V = R/R_t * V_s$$

$$= 3/10 * 20 = 6v = \text{Answer}$$

4: 7v



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ALL LONG QUESTION KVL SAY AY THY SO KVL KI EQUESTION LIKHNA SEKH LO 2, 2, 3,3,5,5 MARKS K SARY QUESTION AVL SAY AY THY PAPER EASY THA

PHY301 MID TERM PAPER SHARED BY STUDENT

ON DECEMBER 13, 2012 AT 8:12AM

past papers main se 2 Mcqs thy

1:current divider rule se circuit solve kran tha

2:aik question main circuit given ha uss ka mesh 1 and 2 ki kvl equation bnti thi

3:circuit given tha KVL equation write krni meah 1 and 2 ki thi

4:matrix equation se solve krna tha aik question 5 marks ka

5:I(not) find karna tha

baqi questions muje yad nhi

theory ka kooi question nhi tha na hi koi past papers main se tha sab circits solve krny thy mcqs bhi solve krny waly thy mostly kvl equation main se kafi paper aya tha

PHY301 MID TERM PAPER SHARED BY STUDENT

ON NOVEMBER 26, 2011 AT 11:32PM

My today's paper of phy301

26 questions were,

MCQs mostly were from past papers

One question was from 22th lecture about matrix----- (5 marks)

and Voltage divider and current divider related questions

also Have to find flowing of current across given resistance



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**Ohms law related questions
and current were given in each question .**

MIDTERM EXAMINATION

Spring 2010

PHY301- Circuit Theory

Time: 60 min

Marks: 40

Question No: 1 (Marks: 1) - Please choose one

If we connect 3 capacitors in series, the combined effect of all these capacitors will be

- ▶ equal to the sum of individual capacitors
- ▶ the sum of reciprocals of individual capacitors
- ▶ zero
- ▶ product of all



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Question No: 2 (Marks: 1) - Please choose one

A node which emerges as a result of combination of two ordinary nodes is called

- ▶ Heavy node
- ▶ Simple node
- ▶ Super mesh
- ▶ Super node

Question No: 3 (Marks: 1) - Please choose one



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Unit of inductance is

- ▶ mho
- ▶ ohm
- ▶ Henry
- ▶ Farad

Question No: 4 (Marks: 1) - Please choose one

Reciprocal of resistance is



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► Conductance

► Inductance

► Capacitance

► Resistivity

Question No: 5 (Marks: 1) - Please choose one

An ideal voltage source is a circuit element where the voltage across it is independent of

► resistance



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► current

► capacitance

► inductance

Question No: 6 (Marks: 1) - Please choose one

Sum of all currents entering in the node is equal to sum of currents leaving the node, is statement of

► Ohm's law

► Kirchhoff's current law



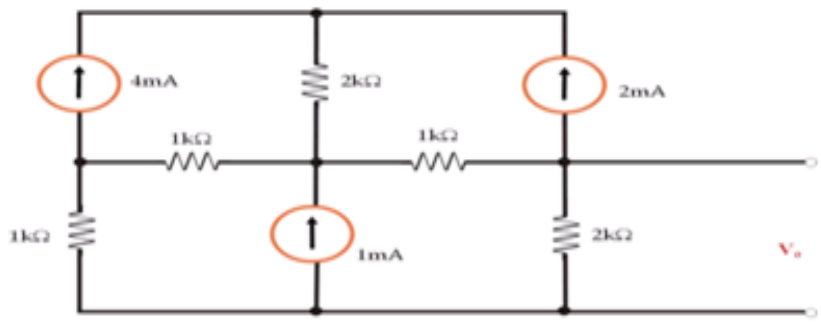
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► Jaime's law

► Rutherford's law

Question No: 7 (Marks: 1) - Please choose one

Which current source will be used for super mesh?



► 4mA

► 2mA

► 1mA

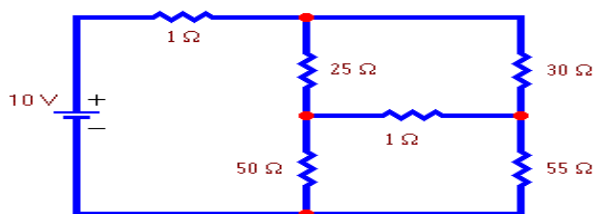


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► 2mA and 4mA

Question No: 8 (Marks: 1) - Please choose one

How many KVL equations can be written for the given circuit?



► 2

► 4

► 3

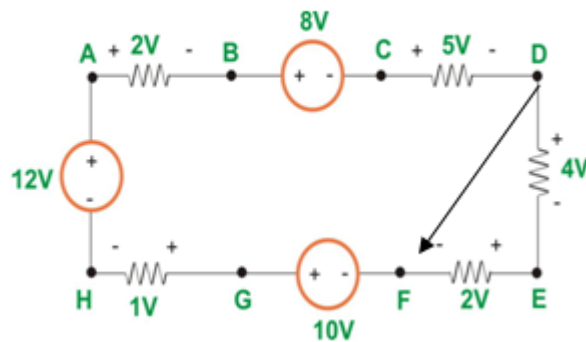
► 1



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Question No: 9 (Marks: 1) - Please choose one

For the given circuit, voltage V_{DF} is



► 6v

► 2v

► 4v

► 10v



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Question No: 10 (Marks: 1) - Please choose one

_____ If a parallel circuit is open in main line, the current

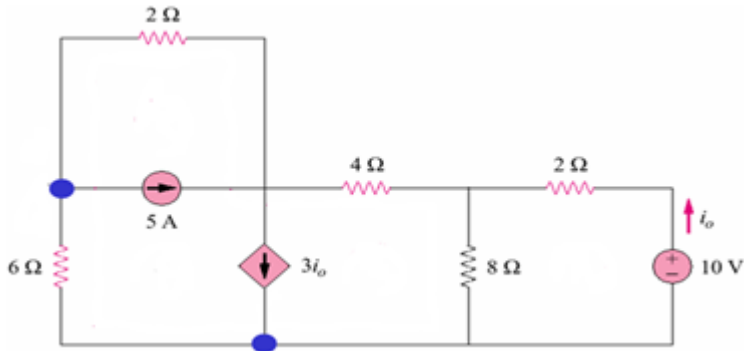
- ▶ increases in each branch
- ▶ is zero in all the branches
- ▶ is zero only in the branch that has highest resistance
- ▶ increases in the branch that has lowest resistance.

Question No: 11 (Marks: 1) - Please choose one



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How many loop equations can be written for this circuit?



► 2

► 4

► 3

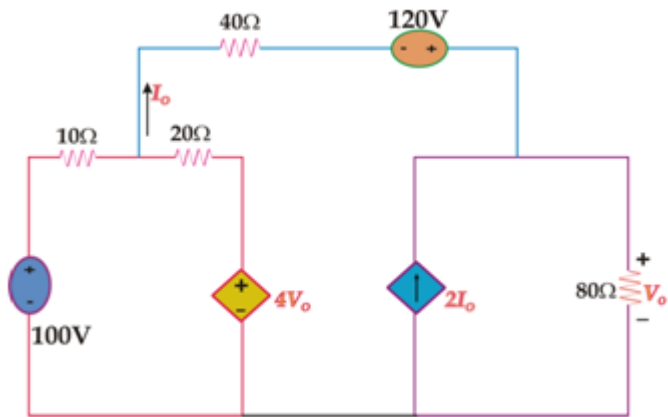
► 5

Question No: 12 (Marks: 1) - Please choose one

How many sources are dependent sources in given circuit?



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► 3

► 2

► 4

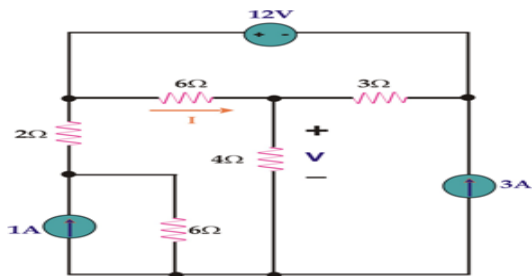
► 1

Question No: 13 (Marks: 1) - Please choose one

How many loops can be assigned to given figure?



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► 2

► 4

► 3

► 5

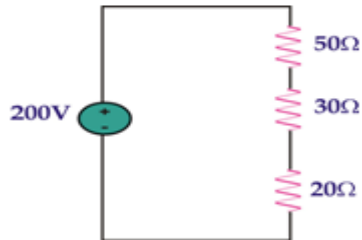
Question No: 14 (Marks: 1) - Please choose one

Through which resistance most current will flow



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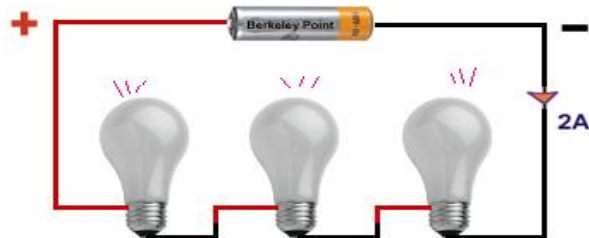
- ▶ 50Ω
- ▶ 30Ω
- ▶ 20Ω
- ▶ same through all resistance

Question No: 15 (Marks: 1) - Please choose one

In given circuit power dissipated across each bulb is 20w.What would be the resistance of each bulb.



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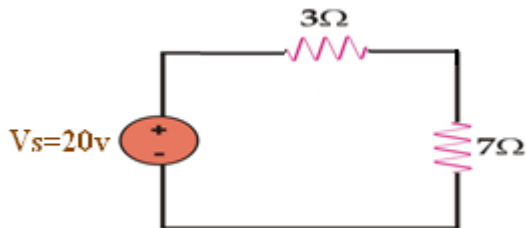
- 5Ω
- 10Ω
- 40Ω
- 20Ω

Question No: 16 (Marks: 1) - Please choose one

In the figure below Voltage drop across 7Ω will be



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► 20v

► 14v

► 6v

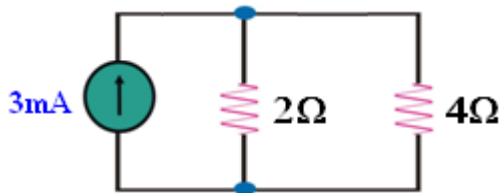
► 10v

Question No: 17 (Marks: 1) - Please choose one

If current flowing through 4Ω is $1mA$, voltage drop across it will be



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► 2V

► 6V

► 4V

► 12V

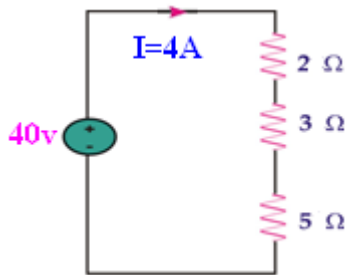
Question No: 18 (Marks: 1) - Please choose one

For the given figure
current flowing through 5Ω is



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► 5A

► 4A

► 10A

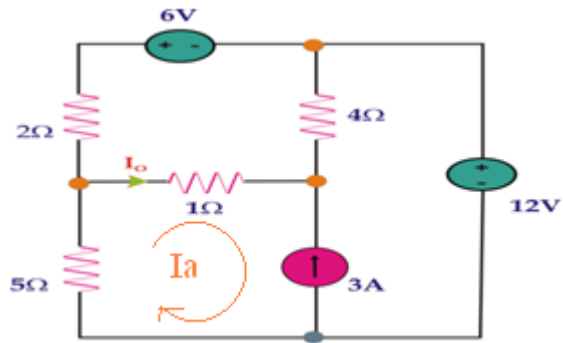
► 2A

Question No: 19 (Marks: 1) - Please choose one

_____ The value of I_a for given circuit is



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► $I_a = I_o$

► $I_a = 3A$

► $I_a = -3A$

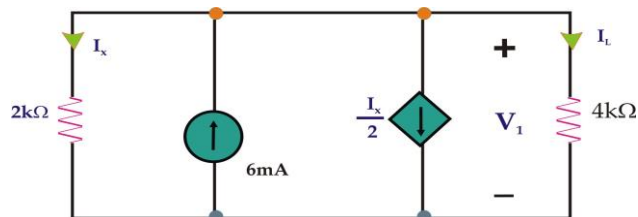
► $I_a = 6v$



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Question No: 20 (Marks: 1) - Please choose one

In the given circuit the value of independent current source is



► I_x

► $I_x/2$



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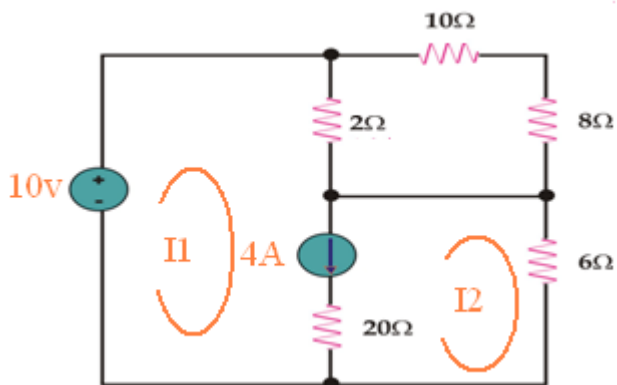
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► 6mA

► IL

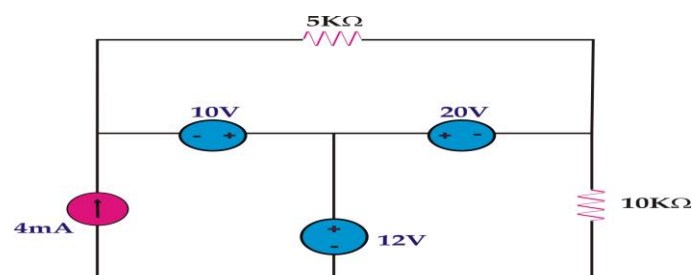
Question No: 21 (Marks: 2)

Write constraint equation for given circuit.



Question No: 22 (Marks: 2)

IDENTIFY AND LABEL EACH NODE FOR THE GIVEN CIRCUIT.



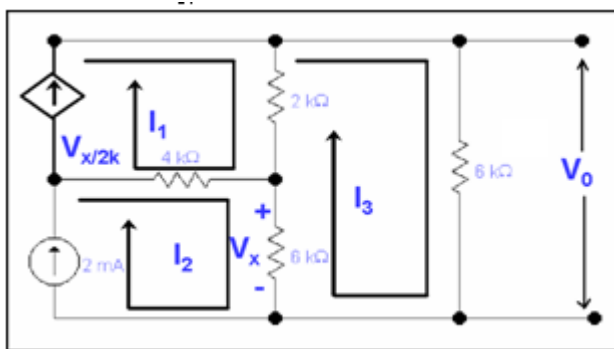


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Question No: 23 (Marks: 3)

Write KVL equations for mesh1 and mesh 3.



Question No: 24 (Marks: 3)

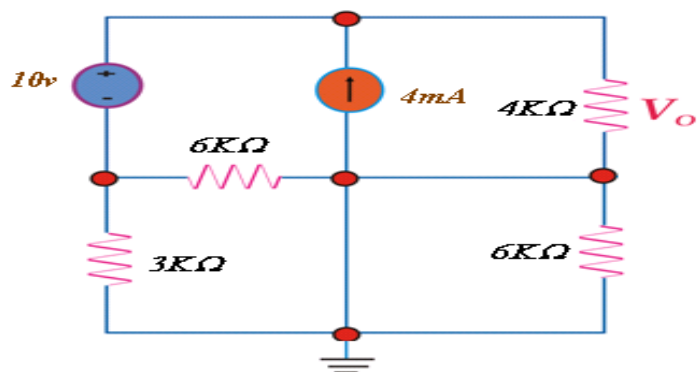




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Question No: 26 (Marks: 5)

Label the nodes and write super node equation for given circuit.



MIDTERM EXAMINATION

Fall 2009

PHY301- Circuit Theory

Time: 60 min

Marks: 33



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Student Info	
Student ID:	Moona
Center:	
Exam Date:	

For Teacher's Use Only									
Q No.	1	2	3	4	5	6	7	8	Total
Marks									
Q No.	9	10	11	12					
Marks									



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Question No: 1 (Marks: 1) - Please choose one

Which of the following statements is correct.

- ▶ **Neutrons and protons lie inside the nucleus.**
- ▶ Protons revolve around the nucleus.
- ▶ Neutron is +ve particle.
- ▶ Electron lies inside nucleus

Question No: 2 (Marks: 1) - Please choose one

Current through a short circuit is



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- ▶ zero
- ▶ minimum
- ▶ **Maximum**
- ▶ leakage current

Question No: 3 (Marks: 1) - Please choose one

When two resistances are connected in series

- ▶ They must both have same resistance value
- ▶ The voltage across each must be the same.
- ▶ They must have different resistance value.

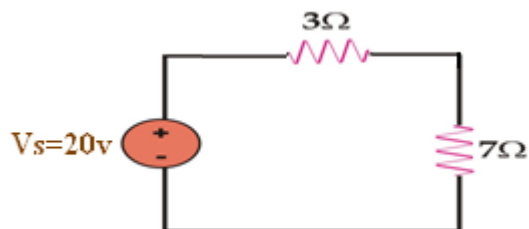


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- There is only one path for current for both resistances

Question No: 4 (Marks: 1) - Please choose one

Using voltage divider in the figure below Voltage drop across 3Ω will be



► 20v

► 14v

► 6v

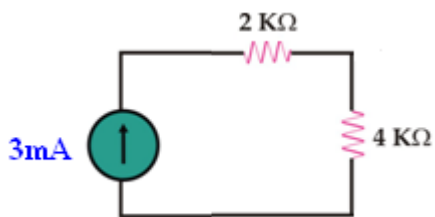


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► 7v

Question No: 5 (Marks: 1) - Please choose one

In the given series fig. the total voltage of circuit is



► 6V



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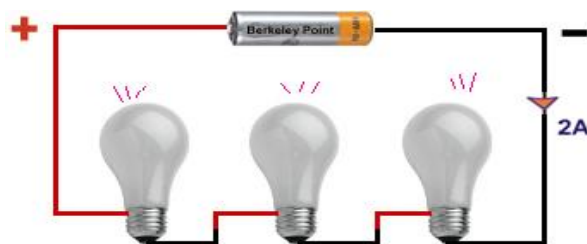
► 12V

► 18V

► 10V

Question No: 6 (Marks: 1) - Please choose one

In given circuit power dissipated across each bulb is 20w. What would be the resistance of each bulb.



► 5Ω

► 10Ω

► 40Ω



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► 20Ω

Question No: 7 (Marks: 2)

What is the unit of capacitance and Inductance

The SI unit of capacitance is the farad; 1 farad = 1 coulomb per volt.

The unit of inductance is the henry (H) named after American scientist and magnetic researcher Joseph Henry. 1 H = 1 Wb/A

Question No: 8 (Marks: 2)

Define KIRCHHOFF'S VOLTAGE LAW (KVL):

Kirchhoff's Voltage Law (or Kirchhoff's Loop Rule) is a result of the electrostatic field being conservative. It states that "the total voltage around a closed loop must be zero". If this were not the case, then when we travel around a closed loop, the voltages would be indefinite. So,

$$\sum V = 0$$

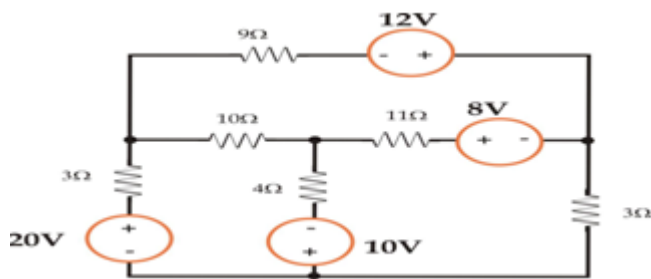


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Question No: 9 (Marks: 3)

Draw and label loop currents for circuit.write kvl for any one loop equation.

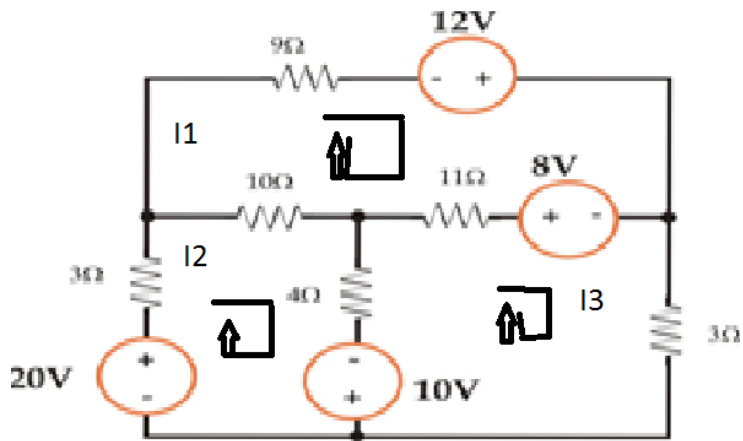


Solution:

Circuit will be redrawn as



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KVL for loop 1 is

Question No: 10 (Marks: 5)

You are given the circuit Use Current Division Rule directly to find I_{30} .

Solution:

Formula is

$$I_1 = R_2 / R_1 + R_2$$



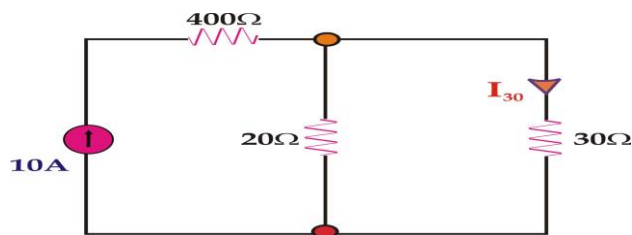
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AS we see that 30 ohm is in parallel with 20 ohm, So the resistance will become 12 ohm..

Now,

Using formula, we have

$$\begin{aligned} I_{30} &= \frac{400}{400 + 12} \times 10 \\ &= \frac{4000}{412} \\ &= 9.7A \end{aligned}$$

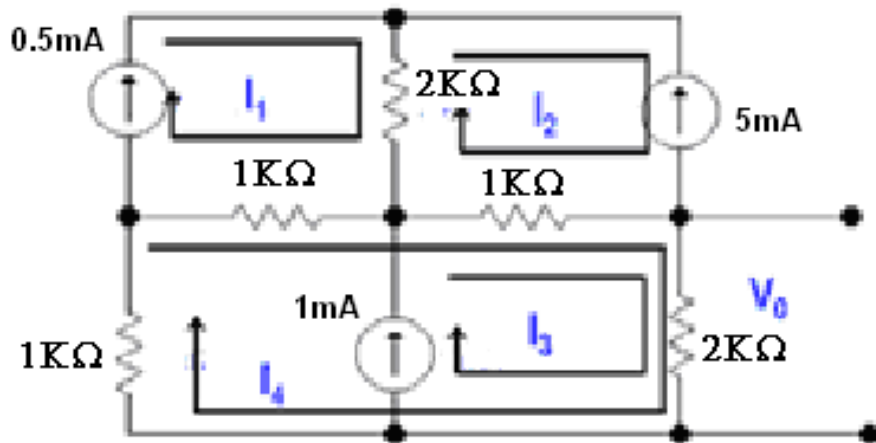


Question No: 11 (Marks: 5)

Determine the value **I₄** for the given network.



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Solution:

$$I_1 = 0.5mA,$$

$$I_2 = -5mA,$$

$$I_3 = 1mA$$

FOR I_4

$$1k(I_4 - I_1) + 1K(I_4 - I_2) + 1KI_3 + 2KI_3 + 1KI_4 = 0$$

$$= 2KI_4 - 1KI_2 + 3KI_3 = 0$$

PUTTING I VALUES

$$= 2KI_4 - 5 + 3 = 0$$

$$= 2KI_4 - 2 = 0$$

$$= I_4 = \frac{2}{2} = 1mA...$$

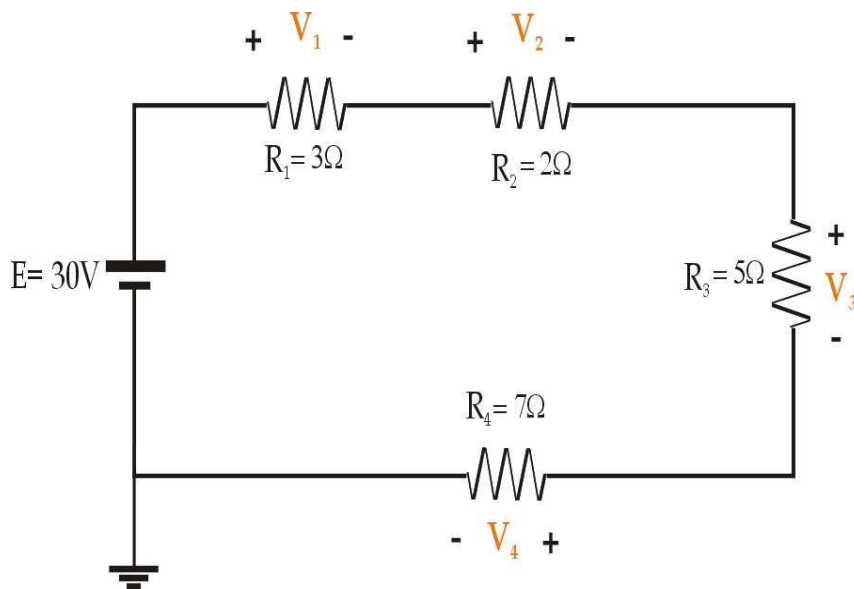
Question No: 12 (Marks: 10)



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If 1.76A current is flowing through circuit

Calculate the power dissipated by R_1 , R_2 , R_3 and R_4 . Determine the power delivered by the source, Write each step of the calculation to get maximum marks and also mention the units of each derived value.





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Solution:

$$I=1.76 \text{ A}$$

As the formula for power delivered by source is

$$P = \frac{V^2}{R}$$

So,

$$P = \frac{30^2}{17} = \frac{900}{17}$$

$$P = 52.94 \text{ Watts}$$

Now Power dissipated in each resistor will



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$$P = I^2 R$$

$$P_1 = (1.76)^2 \times 3$$

$$P_1 = 9.2 \text{ watts}$$

$$P_2 = (1.76)^2 \times 2$$

$$= 6.19 \text{ watts}$$

$$P_3 = (1.76)^2 \times 5$$

$$= 15.48 \text{ watts}$$

$$P_4 = (1.76)^2 \times 7$$

$$= 21.68 \text{ watts}$$